

Lab Title: Steganography and Hidden Data Detection

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Course Name: SBT-DF202: Computer and Digital Forensics

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Computer and Digital Forensics

LAB 4: SBT-DF202: Steganography and Hidden Data Detection

Part A — Prepare and Preserve the Evidence

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ convert -size 400x300 xc:'#4a90d9' \  
-fill white -pointsize 20 -gravity center \  
-annotate 0 'ICDFA Lab Carrier\nBMP for Steghide' \  
BMP3:tower_original_image_for_lab.bmp  
(kali@kali)-[~]  
$ cp tower_original_image_for_lab.bmp tower_original_image_for_lab.bmp.preserved  
(kali@kali)-[~]  
$ file tower_original_image_for_lab.bmp  
tower_original_image_for_lab.bmp: PC bitmap, Windows 3.x format, 400 x 300 x 24, image size 360000, cbSize 3  
60054, bits offset 54  
(kali@kali)-[~]  
$ ls -lh tower_original_image_for_lab.bmp  
-rw-rw-r-- 1 kali kali 352K Sep  4 14:03 tower_original_image_for_lab.bmp  
(kali@kali)-[~]  
$ sha256sum tower_original_image_for_lab.bmp  
3031aabef987421b6f551cf1d23639fbbde30b6c5dcddbc83afca66606949ef4 tower_original_image_for_lab.bmp  
(kali@kali)-[~]  
$
```

The command: `cp tower_original_image_for_lab.bmp tower_original_image_for_lab.bmp.preserved`

Was used to preserve the original image.

The hash below was also generated:

3031aabef987421b6f551cf1d23639fbbde30b6c5dcddbc83afca66606949ef4 tower_original_image_for_lab.bmp

Part B — Create the Secret Evidence File

```
kali@kali: ~/lab4  
Session Actions Edit View Help  
(kali@kali)-[~/lab4]  
$ nano secret.txt  
(kali@kali)-[~/lab4]  
$
```

```
kali@kali: ~/lab4
Session Actions Edit View Help
GNU nano 8.6 secret.txt
Full Name: Samuel Chinedu Onu

This is a controlled ICDFa laboratory secret note for steganography training.

and files within the training environment.
```

HASH RECORD

```
Session Actions Edit View Help

(kali@kali)-[~/lab4]
$ nano secret.txt

(kali@kali)-[~/lab4]
$ sha256sum secret.txt
e23166f26fa3e568bea558a2d3a358e8c15ba6c09ae52dd1d0e6d3face780f03 secret.txt

(kali@kali)-[~/lab4]
$
```

Part C — Embed the Hidden File with Steghide

```
kali@kali: ~/lab4
Session Actions Edit View Help

(kali@kali)-[~/lab4]
$ steghide embed \
  -ef secret.txt \
  -cf tower_original_image_for_lab.bmp \
  -sf tower_stego_lab4.bmp \
  -p 1234 -f
embedding "secret.txt" in "tower_original_image_for_lab.bmp" ... done
writing stego file "tower_stego_lab4.bmp" ... done

(kali@kali)-[~/lab4]
$
```

```

(kali㉿kali)-[~/lab4]
$ sha256sum tower_stego_lab4.bmp
8bf936b6b0c3956f459a04cda0cbf7dfabe539505d63fd6ba24c3c4e6396a466 tower_stego_lab4.bmp

(kali㉿kali)-[~/lab4]
$

```

Part D — Extract and Verify the Hidden Evidence

```

(kali㉿kali)-[~/lab4]
$ mkdir -p extracted

(kali㉿kali)-[~/lab4]
$ steghide extract \
-sf tower_stego_lab4.bmp \
-xf extracted/extracted_secret.txt \
-p 1234 -f
wrote extracted data to "extracted/extracted_secret.txt".

(kali㉿kali)-[~/lab4]
$ sha256sum secret.txt extracted/extracted_secret.txt
e23166f26fa3e568bea558a2d3a358e8c15ba6c09ae52dd1d0e6d3face780f03 secret.txt
e23166f26fa3e568bea558a2d3a358e8c15ba6c09ae52dd1d0e6d3face780f03 extracted/extracted_secret.txt

(kali㉿kali)-[~/lab4]
$ diff -u secret.txt extracted/extracted_secret.txt

(kali㉿kali)-[~/lab4]
$

```

Part E — Compare Original and Stego Images

```

kali@kali: ~/lab4

Session Actions Edit View Help

(kali㉿kali)-[~/lab4]
$ ls -lh tower_original_image_for_lab.bmp tower_stego_lab4.bmp
-rw-rw-r-- 1 kali kali 352K Sep  4 14:03 tower_original_image_for_lab.bmp
-rw-rw-r-- 1 kali kali 352K Sep  4 15:13 tower_stego_lab4.bmp

(kali㉿kali)-[~/lab4]
$ sha256sum tower_original_image_for_lab.bmp tower_stego_lab4.bmp
8031aabef987421b6f551cf1d23639fbbde30b6c5dcddbc83afca66606949ef4 tower_original_image_for_lab.bmp
8bf936b6b0c3956f459a04cda0cbf7dfabe539505d63fd6ba24c3c4e6396a466 tower_stego_lab4.bmp

(kali㉿kali)-[~/lab4]
$ xxd -l 64 tower_original_image_for_lab.bmp
00000000: 424d 767e 0500 0000 0000 3600 0000 2800  BMv~.....6 ... (.
00000010: 0000 9001 0000 2c01 0000 0100 1800 0000  .. .. , , . . . .
00000020: 0000 407e 0500 0000 0000 0000 0000 0000  ..@~.....
00000030: 0000 0000 0000 d990 4ad9 904a d990 4ad9  .... ..J ..J ..J.

(kali㉿kali)-[~/lab4]
$ xxd -l 64 tower_stego_lab4.bmp
00000000: 424d 767e 0500 0000 0000 3600 0000 2800  BMv~.....6 ... (.
00000010: 0000 9001 0000 2c01 0000 0100 1800 0000  .. .. , , . . . .
00000020: 0000 407e 0500 0000 0000 0000 0000 0000  ..@~.....
00000030: 0000 0000 0000 d990 4ad9 904a d990 4ad9  .... ..J ..J ..J.

(kali㉿kali)-[~/lab4]
$

```

The images look visually identical because Steghide uses Least Significant Bit substitution. Changing the least important bit of each colour channel is almost invisible to the human eye, but it completely changes the cryptographic hash. The file size remains the same because the data is written into existing pixel bits.

Part F — Controlled Password Dictionary Test

```
kali@kali: ~/lab4
Session Actions Edit View Help

(kali@kali)-[~/lab4]
$ cat > lab4_wordlist.txt << 'EOF'
password
admin
12345
1234
secret
lab
icdfa
EOF

(kali@kali)-[~/lab4]
$ stegseek tower_stego_lab4.bmp lab4_wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "1234"
[i] Original filename: "secret.txt".
[i] Extracting to "tower_stego_lab4.bmp.out".

(kali@kali)-[~/lab4]
$
```

Part G — Independent Student Task

```
kali@kali: ~/lab4
Session Actions Edit View Help

(kali@kali)-[~/lab4]
$ nano samuel_onu_hidden_note.txt

(kali@kali)-[~/lab4]
$
```

```
kali@kali: ~/lab4
Session Actions Edit View Help
GNU nano 8.6 samuel_onu_hidden_note.txt
Name: Samuel Onu
What is steganography?
Steganography is the practice of concealing a file, message, image, or video within another file so t
Why it matters in digital forensics:
Suspects may hide evidence inside ordinary-looking media files. Forensic examiners must be able to de
```

```
kali@kali: ~/lab4
Session Actions Edit View Help
(kali@kali)-[~/lab4]
$ nano samuel_onu_hidden_note.txt
(kali@kali)-[~/lab4]
$ sha256sum samuel_onu_hidden_note.txt
fc4e210ee40b024730bc1c4cc5ba79fe1d12c24f75a2fbed0c240ddee17b3f9c samuel_onu_hidden_note.txt
(kali@kali)-[~/lab4]
$
```

1. Embed it into the carrier image.

```
kali@kali: ~/lab4
Session Actions Edit View Help
(kali@kali)-[~/lab4]
$ steghide embed \
  -ef samuel_onu_hidden_note.txt \
  -cf tower_original_image_for_lab.bmp \
  -sf lab4_stego_independent.bmp \
  -p lab4password -f
embedding "samuel_onu_hidden_note.txt" in "tower_original_image_for_lab.bmp" ... done
writing stego file "lab4_stego_independent.bmp" ... done
(kali@kali)-[~/lab4]
$
```

Password used is **lab4password**

3. Extraction

```
(kali㉿kali)-[~/lab4]
└─$ steghide extract \
    -sf lab4_stego_independent.bmp \
    -xf extracted_independent/recovered_note.txt \
    -p lab4password -f
wrote extracted data to "extracted_independent/recovered_note.txt".

(kali㉿kali)-[~/lab4]
└─$
```

4. Verify hash

```
(kali㉿kali)-[~/lab4]
└─$ sha256sum samuel.onu_hidden_note.txt extracted_independent/recovered_note.txt
sha256sum: samuel.onu_hidden_note.txt: No such file or directory
fc4e210ee40b024730bc1c4cc5ba79fe1d12c24f75a2fbed0c240ddee17b3f9c  extracted_independent/recovered_note.txt

(kali㉿kali)-[~/lab4]
└─$ diff samuel_onu_hidden_note.txt extracted_independent/recovered_note.txt

(kali㉿kali)-[~/lab4]
└─$
```

5. Word list creation

```
(kali㉿kali)-[~/lab4]
└─$ cat > student_wordlist.txt << 'EOF'
password
1234
Qwerty
MyLabPass
MyLabPass2026
lab4password
EOF

(kali㉿kali)-[~/lab4]
└─$ stegseek lab4_stego_independent.bmp student_wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "lab4password"
[i] Original filename: "samuel_onu_hidden_note.txt".
[i] Extracting to "lab4_stego_independent.bmp.out".

(kali㉿kali)-[~/lab4]
└─$
```